



Readdar:

Where Books Spark Connections

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CPT208 Human-Centric Computing
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Introduction

Project Domain: Generative AI in **Deep Reading & Chatting Assistance**

Introduction: Radar is an AI-powered platform that **connects readers for one-on-one, topic-focused discussions**. By analyzing reading behavior and annotations, it matches users and promote thoughtful, structured conversations. Turning reading into a richer, more interactive experience.

Problems to Address⁰¹

Discussions on public platforms lack of depth

Online book discussions often become shallow or redundant, missing structured logical progression.

Finding a suitable discussion partner is difficult

Reading the same book does not guarantee intellectual compatibility. Readers need macro-level idea alignment, not just shared book titles.

Starting a discussion is daunting

Many readers struggle with starting or sustaining deep discussions, especially in environments where their viewpoints might be misunderstood or challenged aggressively.

Diff from Existing Works⁰¹

Traditional social platforms & reading communities

(e.g., Douban, Goodreads): discussions are mostly group interactions, easily lead to pandering and disengagement.

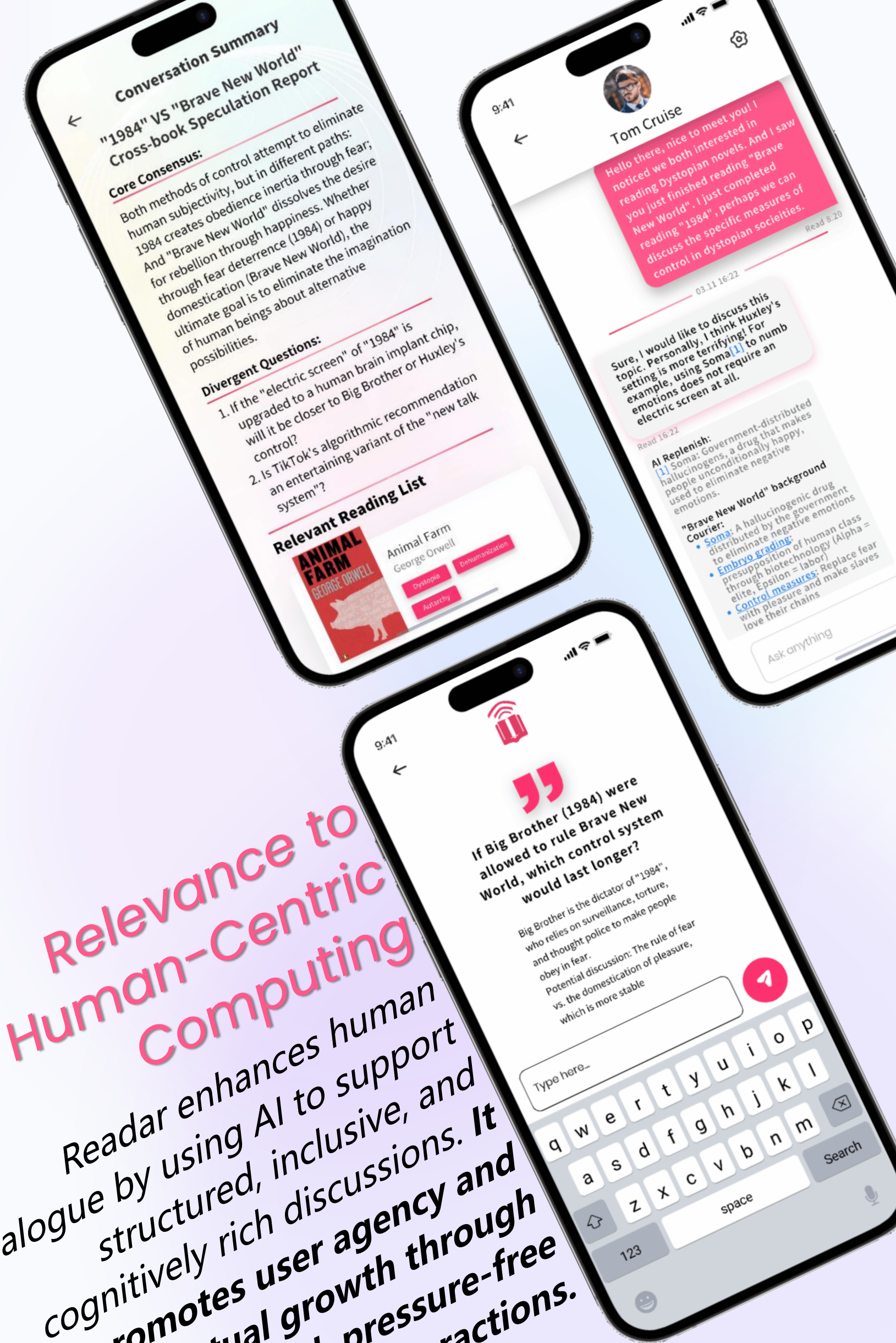
Personalized reading tools

(e.g., Calibre, Koodo): Some support annotation or sharing but lack targeted discussion matching and AI-guidance. Fail to promote a true exchange of opinions.

Generative AI dialogue system

(e.g., ChatGPT, Deepseek): Focus on Q&A but does not focus on the multi-dimensional interaction and logical extension of in-depth reading and discussion.

Relevance to Human-Centric Computing
Radar enhances human dialogue by using AI to support structured, inclusive, and cognitively rich discussions. **It promotes user agency and intellectual growth through personalized, pressure-free interactions.**



User Persona



XIAOYANG LIN
IT Engineer

Gender : Male
Age : 25
Location: Shenzhen
Status : Busy Working

“There’s NO forced communication, just because we are both interested in it.”

> Fragmentary Time > Restricted by work

BIOGRAPHY

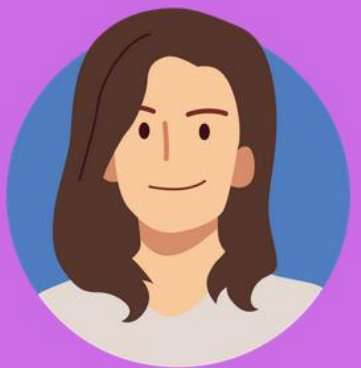
Lin Xiaoyang is an IT Engineer who reads in his spare time. Due to **busy working schedule**, he found it difficult to spend time socializing in offline book clubs.

He seeks inspiration and deeper thinking but **lacks suitable discussion partners**. He aims to connect with diverse people through various topics and organize fragmented knowledge.

GOALS & NEEDS

- Use intelligent tools to organize fragmented knowledge
- Enhance the depth of discussion
- Communicate with people from different backgrounds

FAVORITE BOOKS



JANE WANG
International Student

Gender : Female
Age : 21
Location: England
Status : Busy in Lab

“Seeking minds to collide with — accross time zones and ideas”

> Time Difference > Lack of Social Partners

BIOGRAPHY

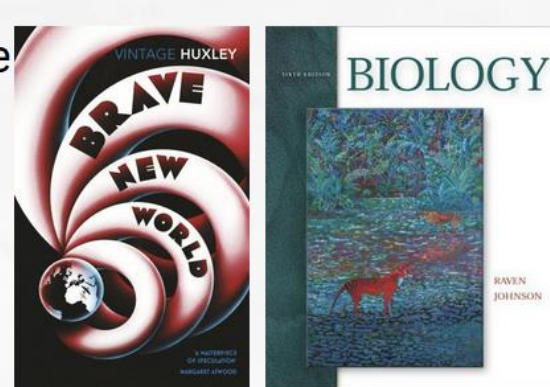
Jane Wang, an international biology student, is **too busy with lab experiments** to find local readers to share ideas with. Due to the **time difference**, communicating with people in her home country is also challenging.

She aims to find someone to exchange thoughts with and to have fruitful exchanges with the assistance of GAI.

GOALS & NEEDS

- Find a partner who is acceptable in time
- Utilizing GAI to achieve a more productive communication
- Finding quality discussions with people eager to share ideas

FAVORITE BOOKS



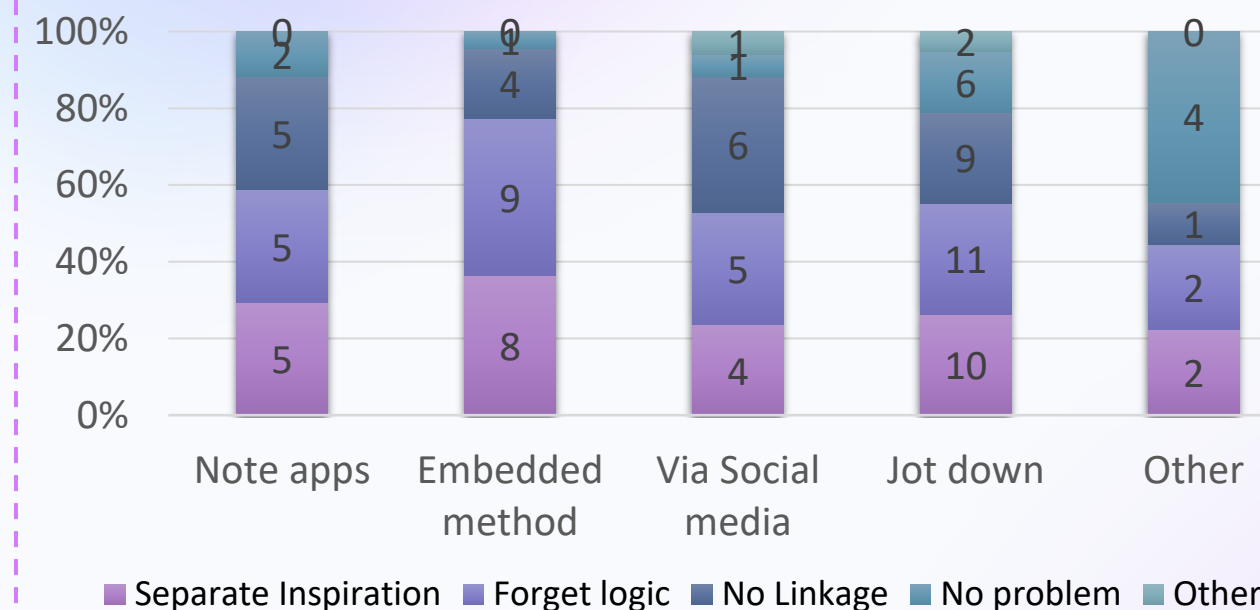
Similarities: Users have reading habits and want to have in-depth discussions, but constrained by their environment (time, social barrier, output ratio, etc.) to do so.

Key Findings of Requirement

Click [Here](#) to check out How we conducted the survey [1] and the **target group** of data.



Cross analysis of Reading difficulties and Note taking methods



01 Need of Integrated Insight Capture and Management

Most users (59%) record ideas informally, while little (30%) use in-app annotations. Though 62% agree annotations aid understanding, only 24% highlight.

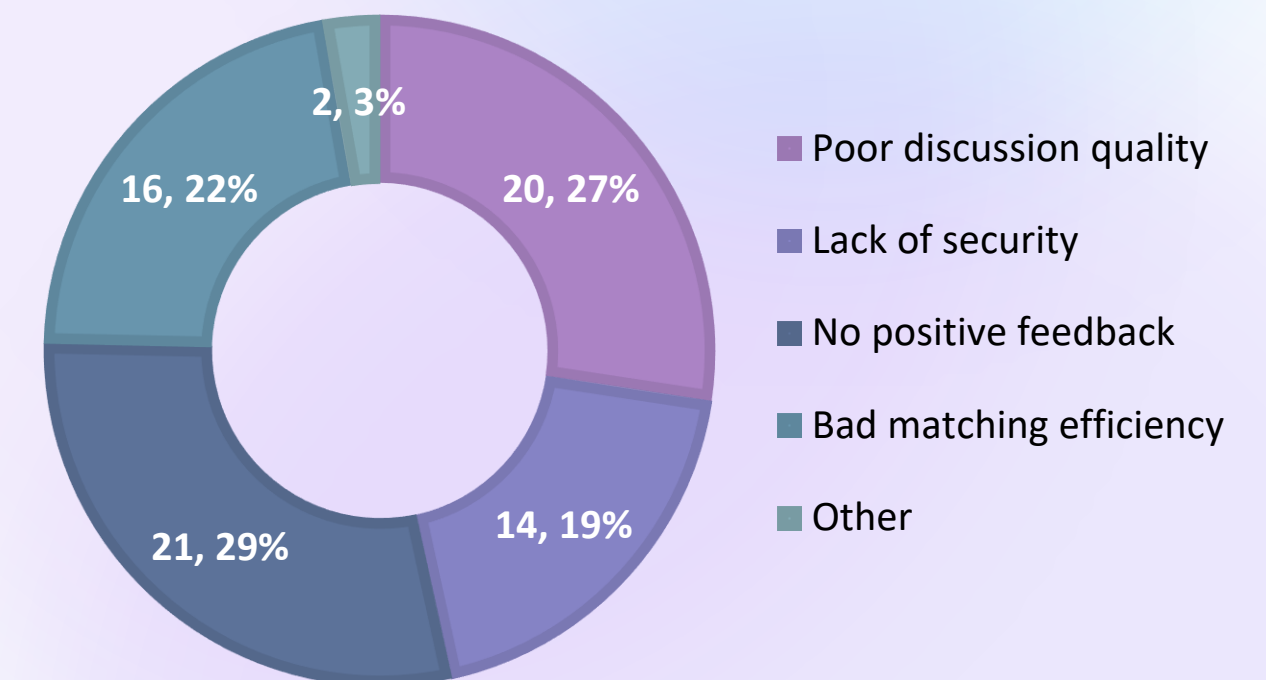
➤ **Users lack systematic tools to transform spontaneous inspiration into structured knowledge.**

02 Desire for Discussion vs. Social Barriers

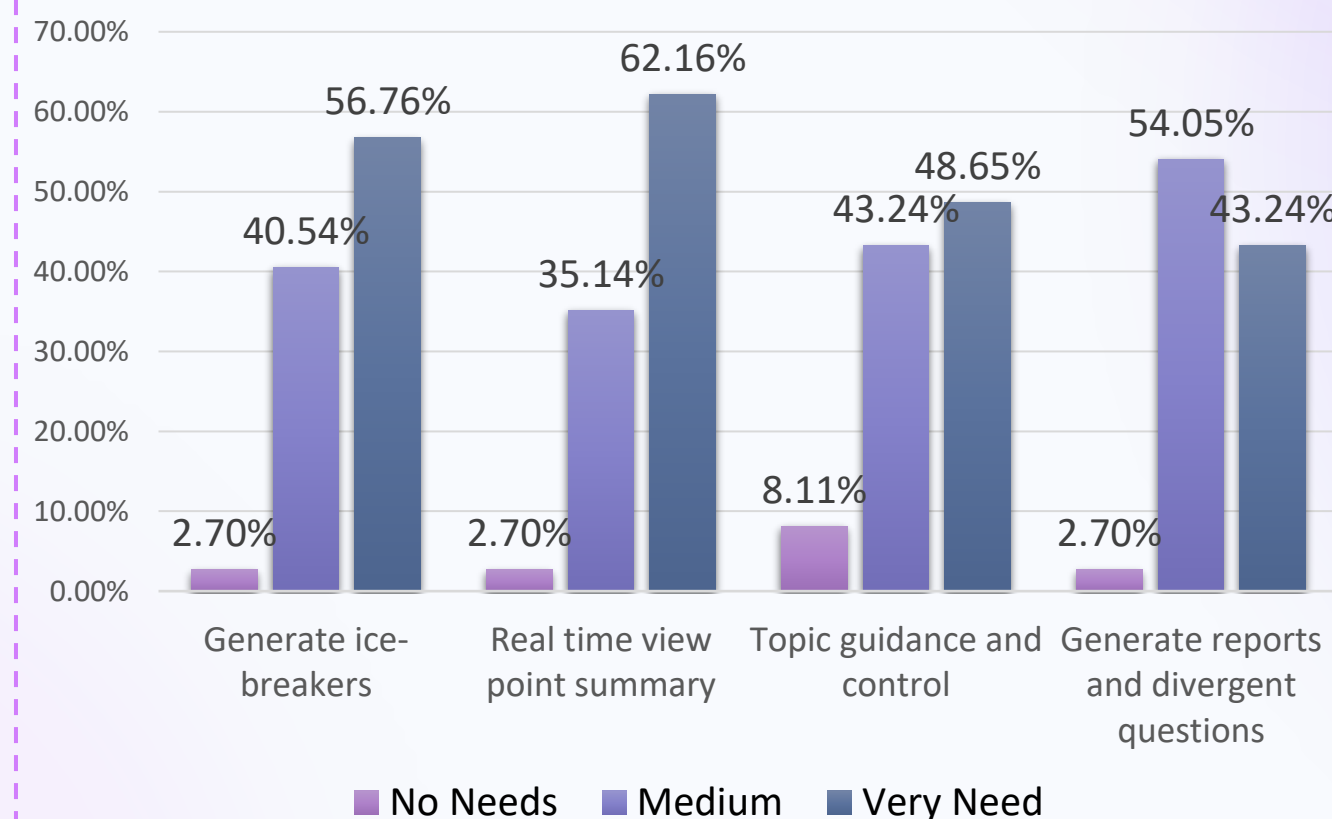
While 51% wish to discuss resonant content, 84% fail due to mismatched reading experiences.

➤ **Inefficient matching mechanisms and unsafe discussion environments blocked meaningful exchanges.**

Core Obstacle to Deep Communication in Existing Reading Communities



User's interest in AI functionality



03 GAI-Driven Conversation Support

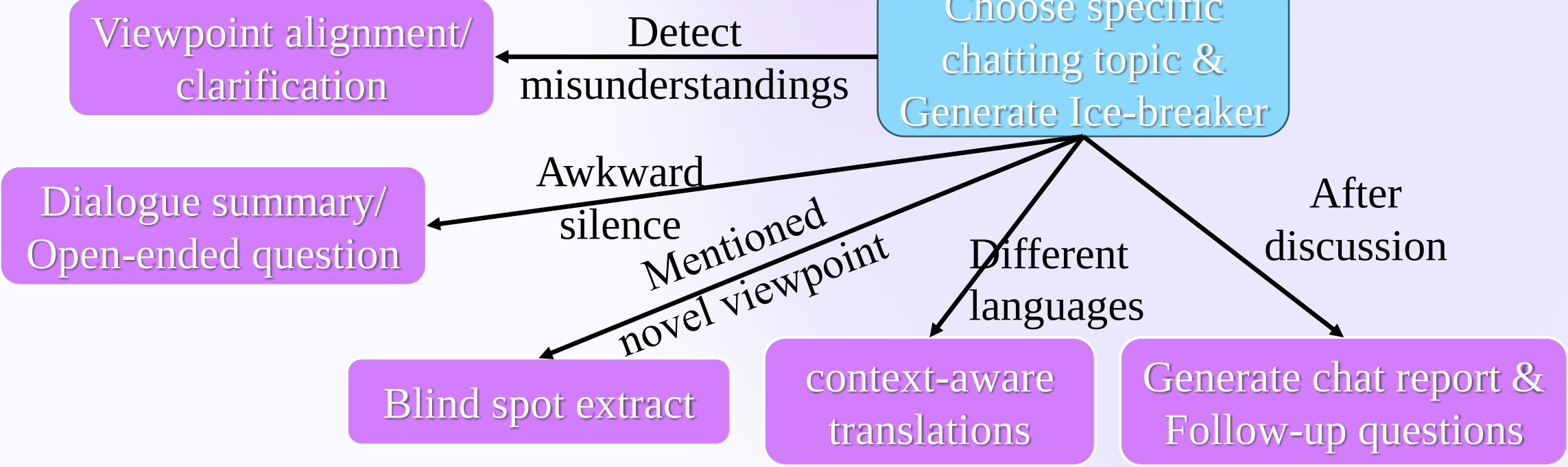
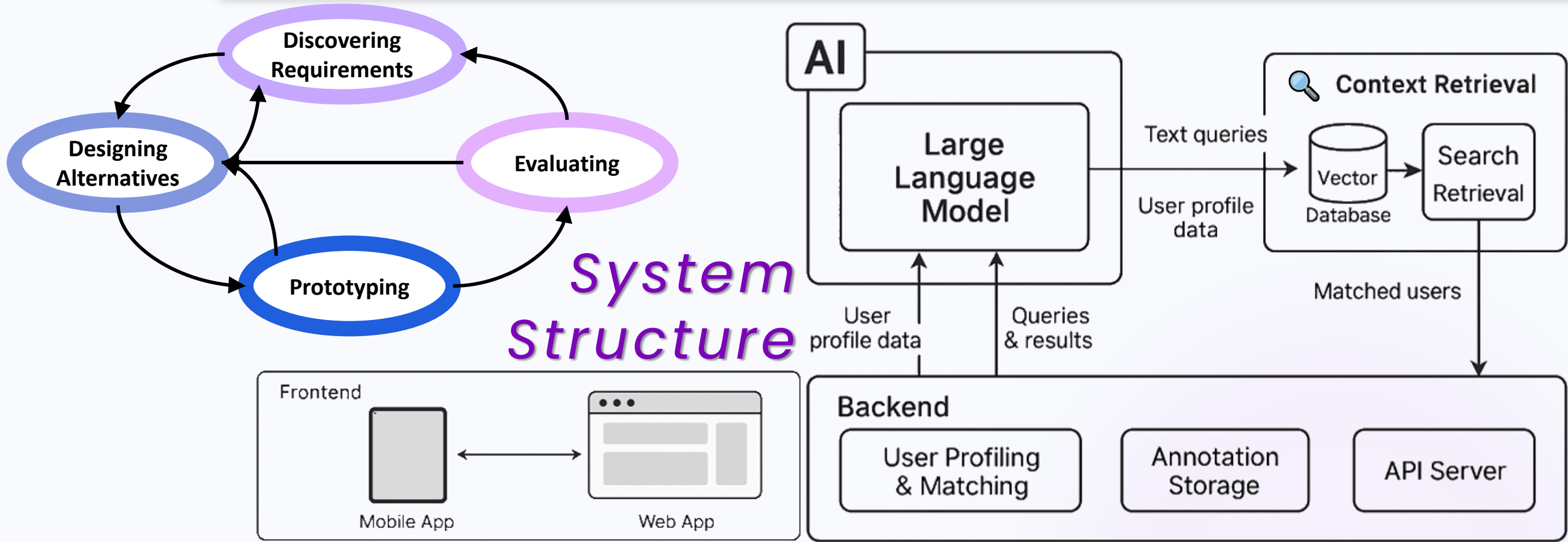
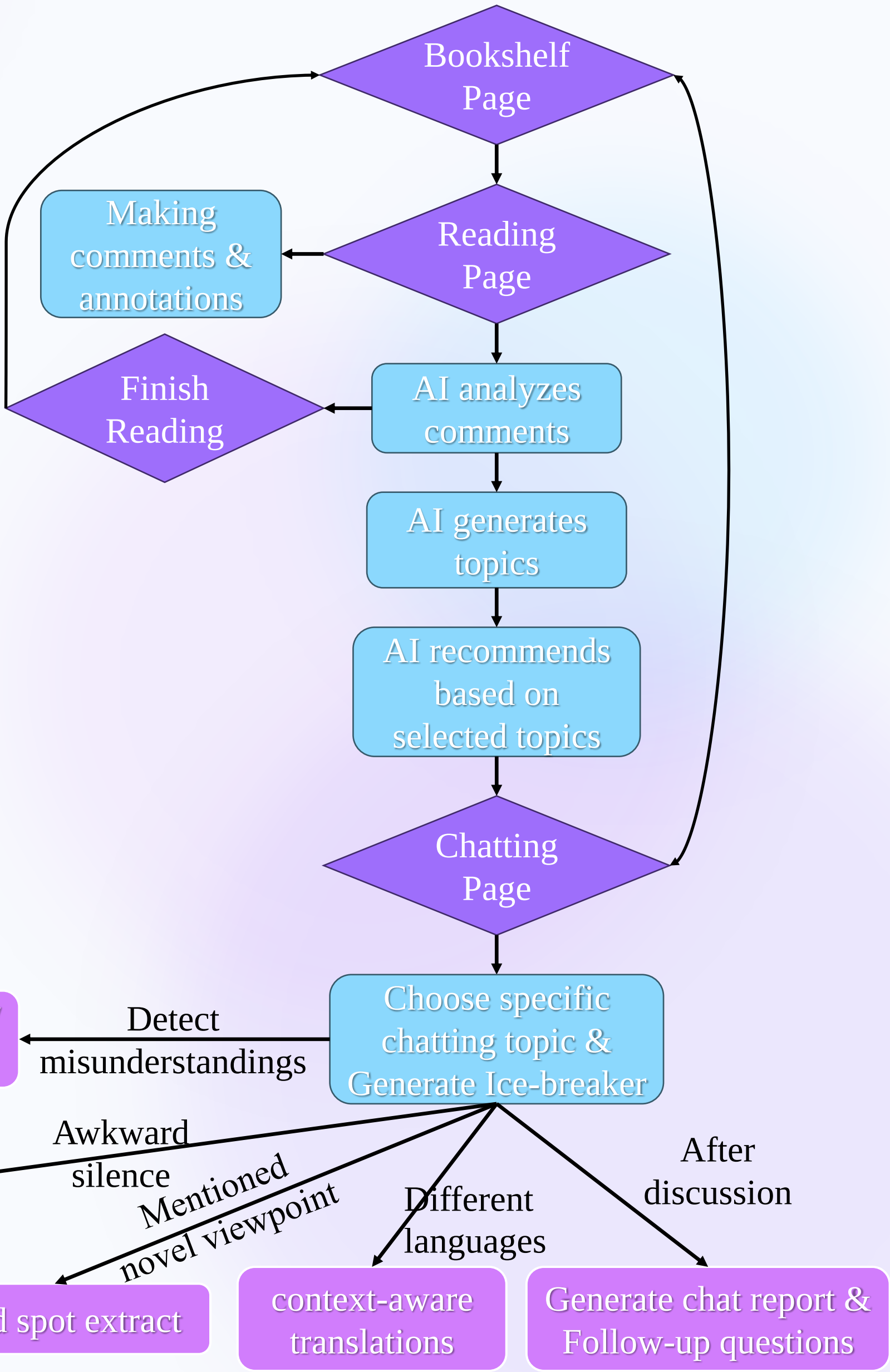
Users Valuing AI features like icebreaker generation (57%) and in-text quote support (76%).

➤ **There is significant interest in AI functionalities and most user hope GAI to promote more insightful chats while ensuring privacy.**

Iteration Table

Function Iteration	Key information collection	AI matching mechanism	AI-assisted conversation
Iteration I	Collect readers' comments and annotations; Record reader's reading time and mood fluctuation via wearable device.	Search users who recently read the same book and written similar comments; User with similar time routine and experience.	AI generate icebreaker topics about overlapping viewpoints; Present corresponding comments during conversation;
Iteration II	Remove the mood fluctuation recording; Record the reader's annotations and viewpoints.	Expands matching scope to books on the same subjects; AI focuses on matching viewpoints in comments.	AI summarizes the plot in the book read by another reader; generates real-time chat summaries and follow-up topics.
Iteration III	AI records the key words that mentioned multiple times in comments.	Search who shows interest in similar topics;	AI adds online search function; Showing Notice board when someone misunderstood another's view.
Iteration IV	User's book content, corresponding jargon and dialogue context.	Match users across different languages who share overlapping interests; Prioritize cultural diversity.	AI generate context-aware (dynamically fine-tuned) translations.

Flow Chart



Requirements

Interview: A semi-structured many-to-one interview [1]

Focused on: Reading habit; Annotation preferences; discussion experiences; expectations from AI assisted reading tools.

Key Results:

- ✓ Need of in-depth discussion after reading.
- ✓ Encourage the expression of controversial or non-traditional ideas.
- ✓ Expect similar thinkers based on ideas and value for discussion.
- ✓ Lower the psychological and operative threshold for initiating discussions.
- ✓ Quick access to citation support for the original article or relevant comments during discuss.

Design

Software format: Plug-in for Reading software
Using permissions collecting reading duration, annotated content, mood swings (e.g., repeated page turns, pause duration, heart rate activity).

User matching & discussion design:

- Extract annotated text topic vectors;
- Initial matching based on topic similarity and activity;
- Ice-breaking questions for both parties' annotations are generated by AI before the chat starts.

Evaluation

User feedback:

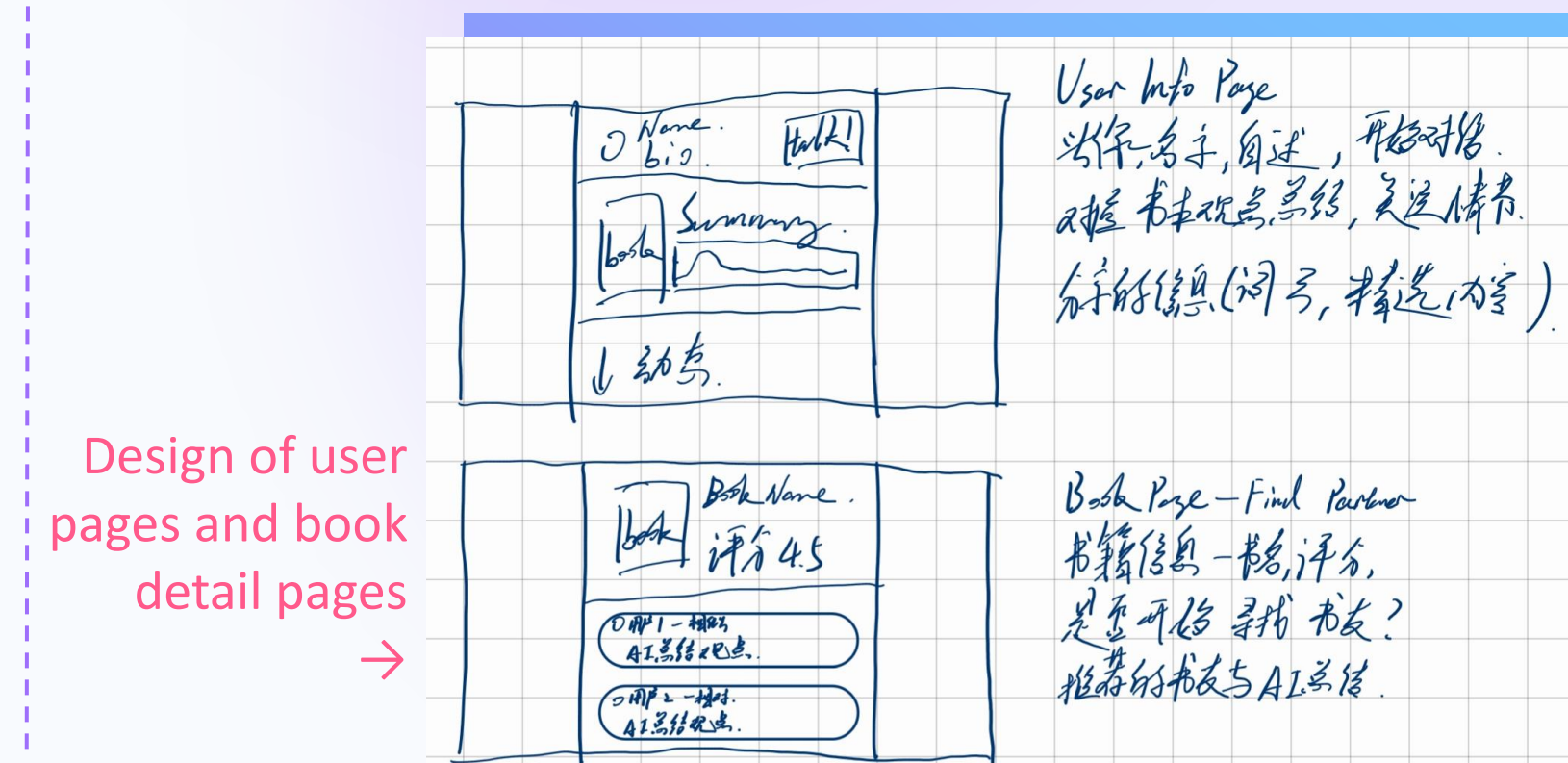
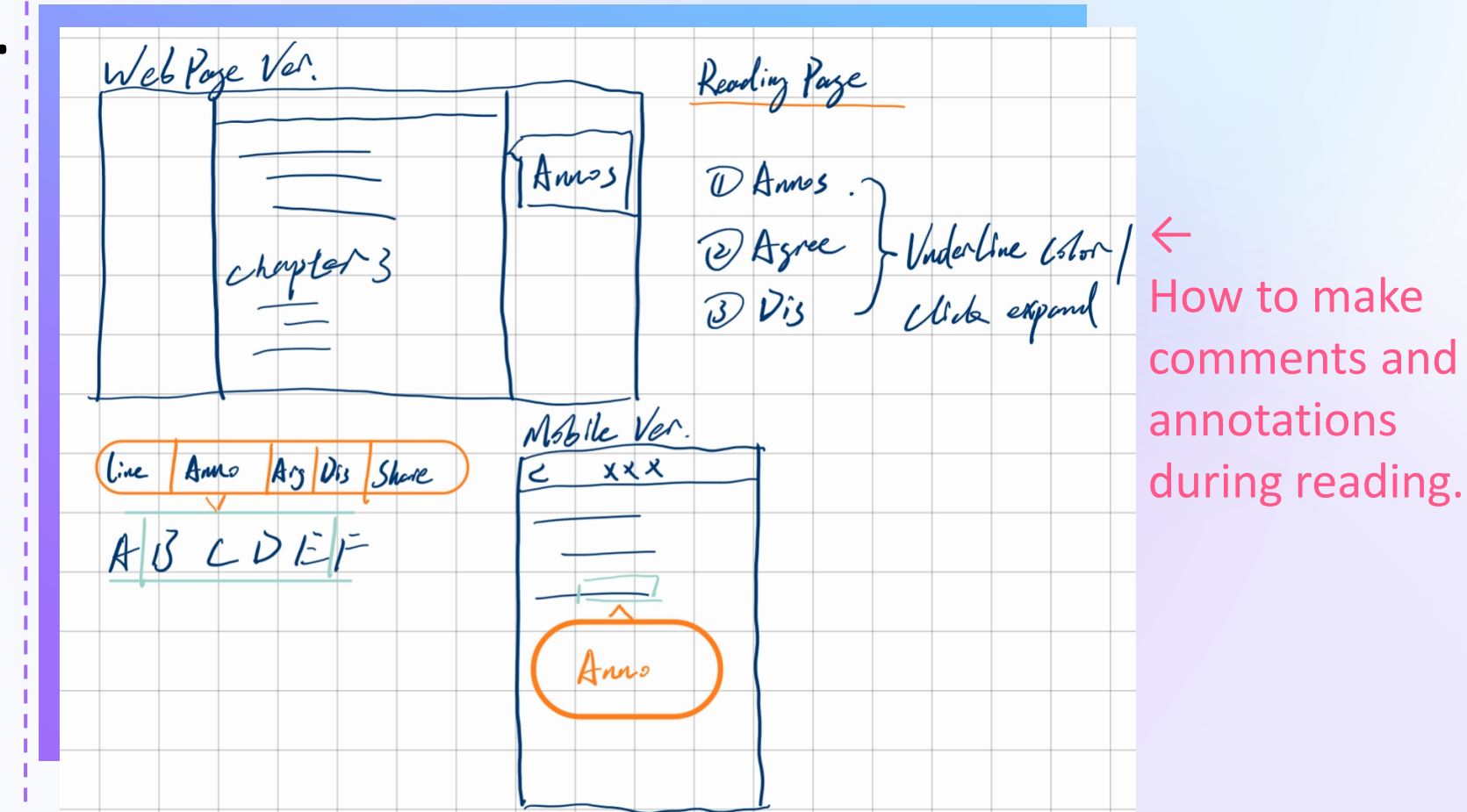
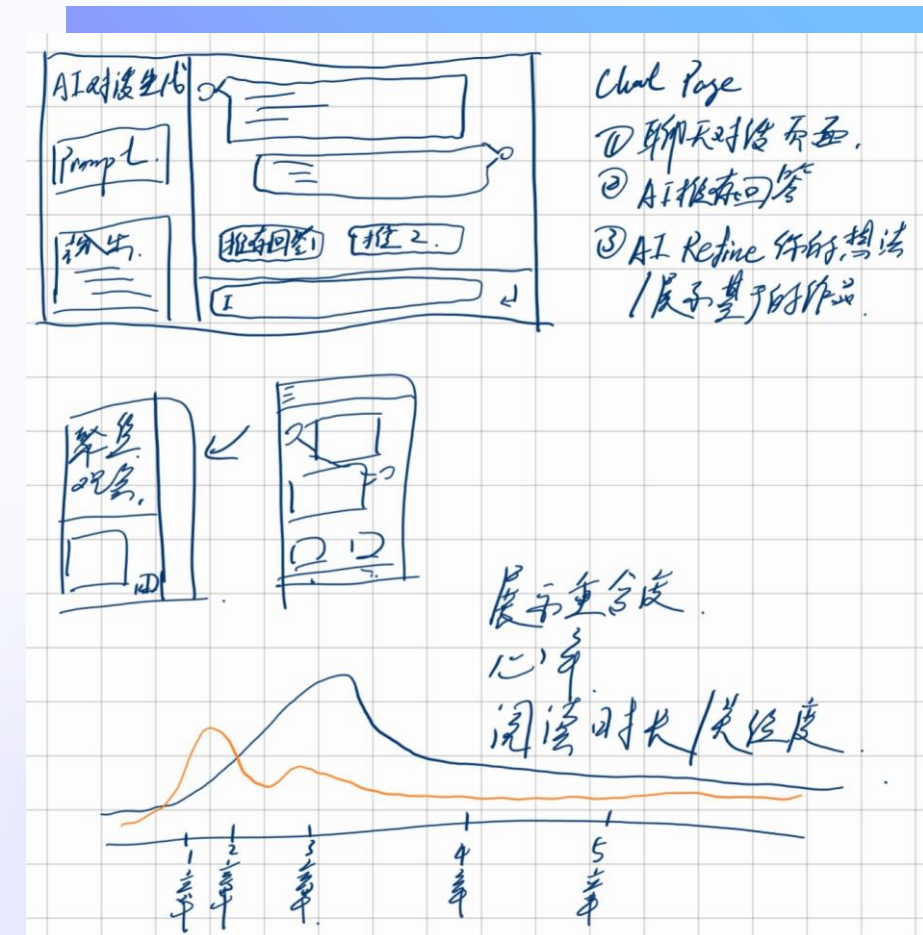
- ❑ Plug-in is too intrusive on privacy and are not willing to share emotional data;
- ❑ Separation of chat entrances make it hard to initiate discussions at will;
- ❑ Poor matching of niche books due to too small matching pool.

Reflection:

- ❑ Proof of concept was successful and confirmed core pain points;
- ❑ Expand the matching range and lowering the user threshold.

Prototype (conceptual)

AI-assisted discussion and matching design →



Requirements

Survey: An online questionnaire [1], two related survey reports [2], [3].

Focused on: Reading behavior, annotation practices, discussion habits, and attitudes toward AI functionalities.

Key Results:

- ✓ Expanding the scope of matching: 56.75% of users [1] found it difficult to start because of the difficulty of matching cold books .
- ✓ Demand for assisting discussion: 62% of users [1] think real-time point of view summary and mind map are helpful for chatting; 49% [1] expect AI to provide guidance, such as topic guidance, preventing cold scenes, and so on.
- ✓ Demand for follow-up discussions: More than half of the users [1] were interested in generating chat reports and follow-up questions, hoping would facilitate potential secondary discussions.

Design

Software format: standalone mobile App: integrated reader and discussion function;

User matching & discussion design:

- Improved Chat Support: New features such as quote support, summary of views, and question follow-up have been added to the chat support design.
- Change matching mechanism : matching of the same book → matching of the same kind of topic that the readers are interested in.

Evaluation

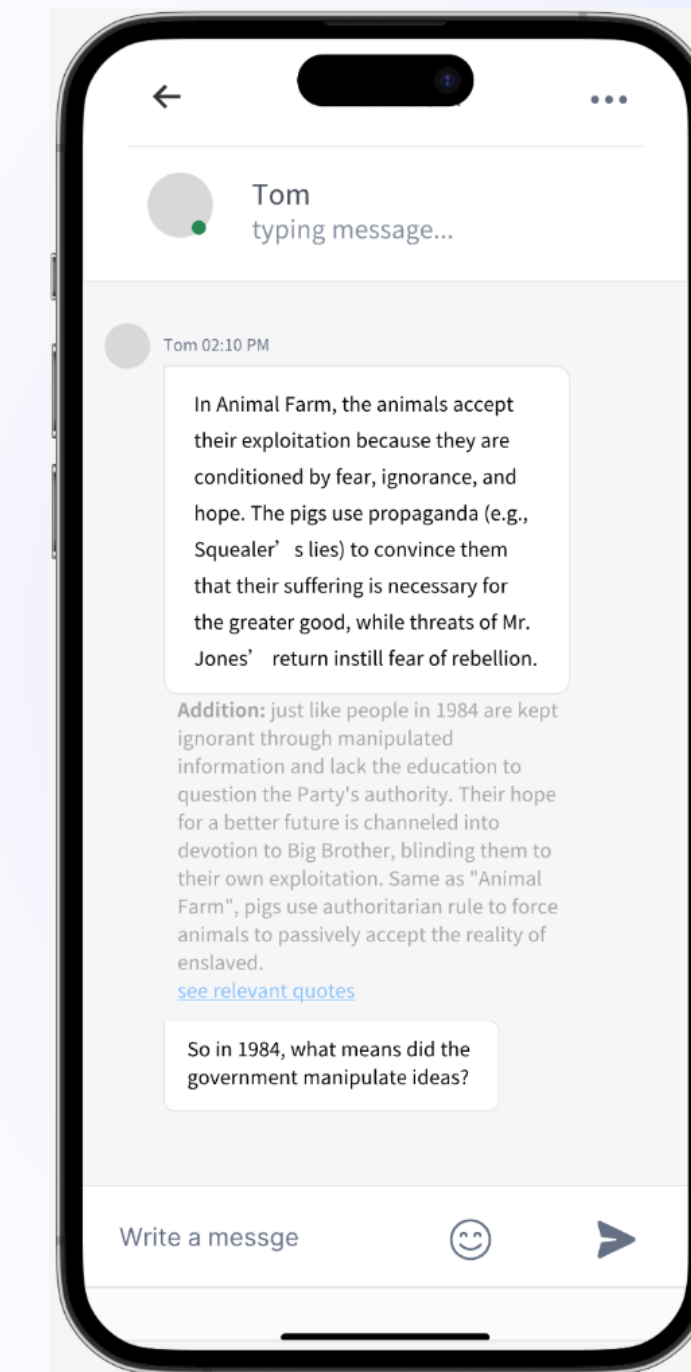
User feedback:

- ❑ Users are satisfied with the expanded match range and the new assist features.
- ❑ Over 70% participants said real-time viewpoint mapping helped clarify thinking.
- ❑ Matching of readers to niche books rises.

Reflection:

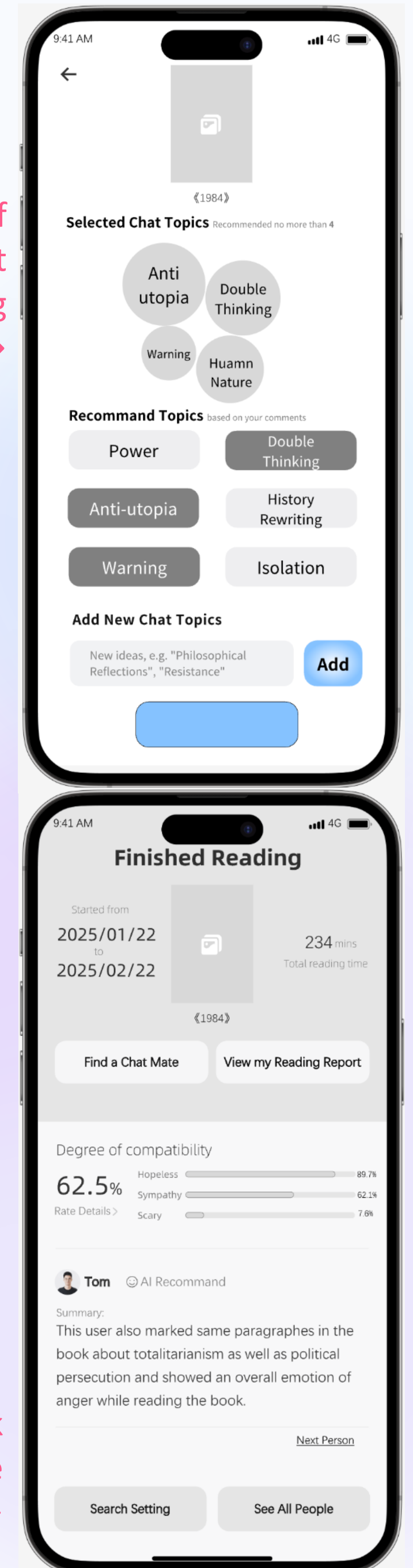
- ❑ Need for more intuitive and clearer UI/UX design.
- ❑ Need ai to support alignment of viewpoints when misunderstandings arise.

Prototype



↑ Real-time Blind spot extract during discussion generated by GAI

Design of Book Details Page →



Requirements

- ✓ **Information and error correction:** Users expect AI to provide additional information that helps them understand the perspective of the person they are communicating with, so that they can better communicate in depth. (38% of users chose not to engage in exchanges after reading because they were afraid being misinterpreted [1])
- ✓ **Efficient and Accurate Discussions:** Users need AI to have the ability to screen information and avoid topic deviation through guidance to ensure efficient and accurate discussions.
- ✓ **Cross-boarder communication:** Many users would like to have the opportunity to communicate with each other to broaden their horizons in different fields.

Design

Online search with 'bulletin boards':

Integration with external search interfaces (e.g., wikis, related websites) for quick retrieval of contextual information when unfamiliar content exists;

Better matching mechanism:

Based on topic tag clustering, matching readers across book topics;

Interest Extraction and Analysis:

AI records keywords that are repeatedly mentioned in comments and analyses the topic areas that users are interested in.

Evaluation

Heuristic Evaluation feedback:

Evaluated by 4 experts in the field of reading, recommended further optimization of the prompting strategy to reduce disruption.

User feedback:

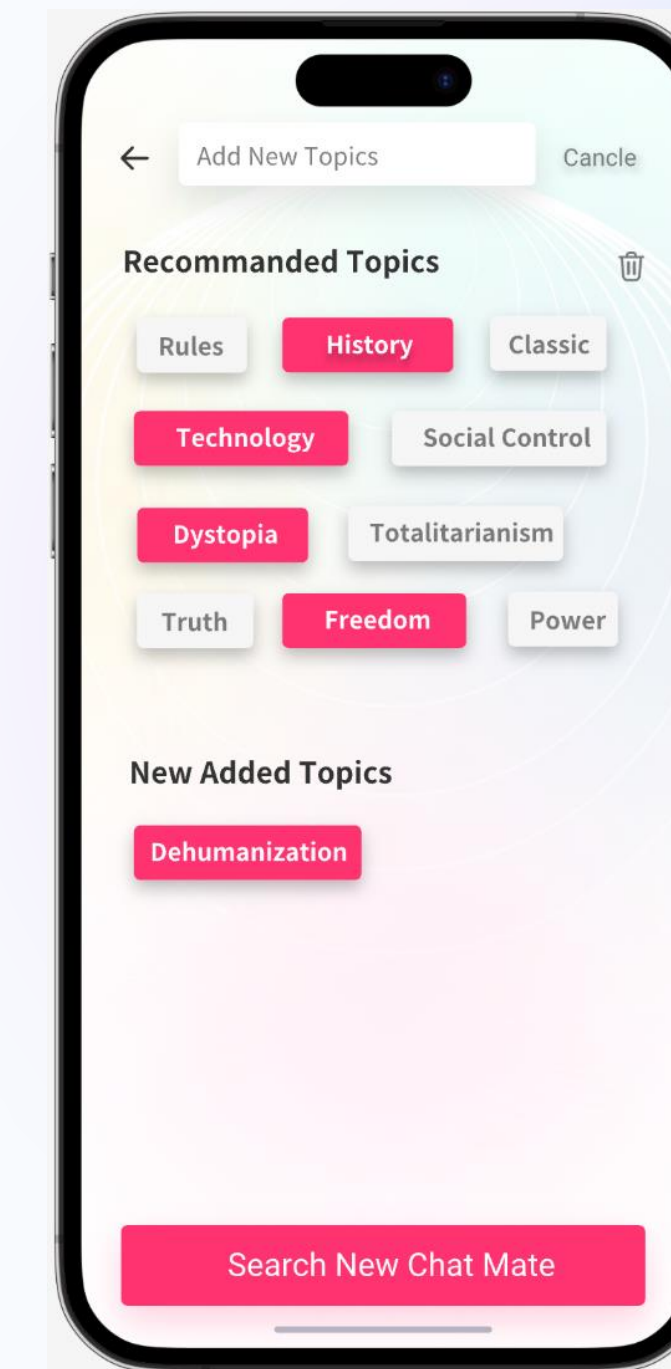
Through user interviews, 75% of users believe that the new features made the process richer.

Reflection:

- ❑ Optimize ai presentation strategy, Cognitive load may occur when there are too many keyword cues.
- ❑ Get a wider range of views and ideas while expanding the user base.

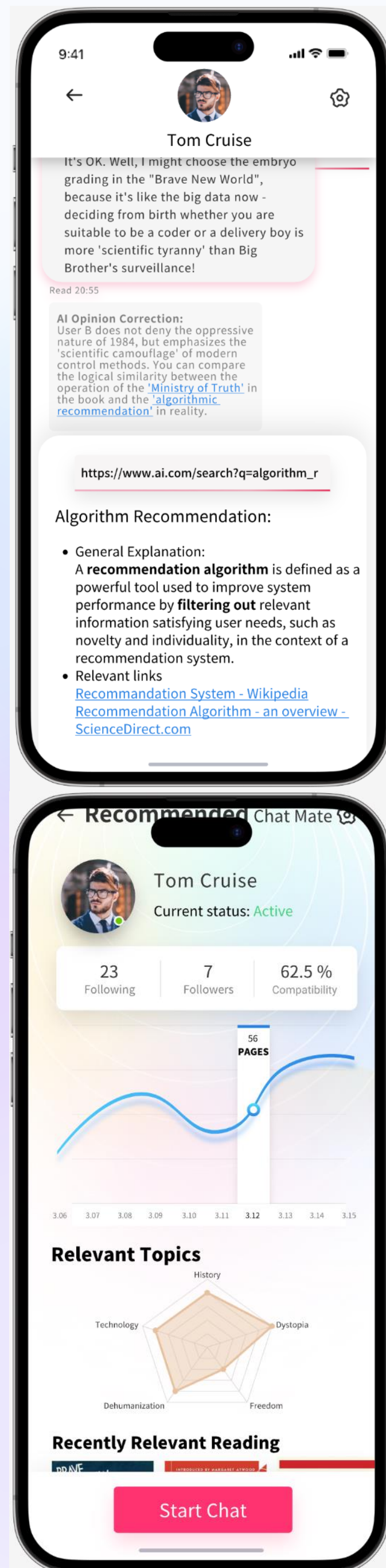
Prototype

Embedded Information and error correction



↑ User interest extraction and topic selection.

Design of User detailed info page



Requirements

- ✓ **Dialogue Summary and Follow Up:** After a discussion, users expect to generate dialogue reports and derived questions [1],[3] → facilitate follow-up exchanges and maintain the coherence and depth of the discussion.
- ✓ **Multilingual intercultural discussion:** Users want to communicate with exotic readers to engage cross-cultural perspectives → support automatic, context-aware translation to avoid mistranslation and semantic drift.
- ✓ **Selecting Chatting Partner:** After selecting a topic, users would like to view the thoughts of different readers to select chatting partners according to their preferences → improving the quality of communication and matching interests.

Design

Dialogue Report and Derivative Question Generation:

At the end of the chat, the AI generates a dialogue report, including summary of both views, highlights and unresolved issues.

Based on the dialogue content and the interests, AI generates derived questions.

Integration of AI Translation:

Embedded in chat interface, translate in real time and dynamically adjust the result in the context of the dialogue and book content to improve the accuracy and consistency.

Partner selection:

Displays the summary of readers' ideas, opinion tendency and discussion history of the topic.

Evaluation

A/B Test (N=20):

- **Experimental group:** new context-aware translation + AI bootstrapping;
- **Control group:** using regular translation + no bootstrapping;

Users with optimized features scored 48% higher on satisfaction with cross-language communication and communication duration was 31% longer on average.

User feedback:

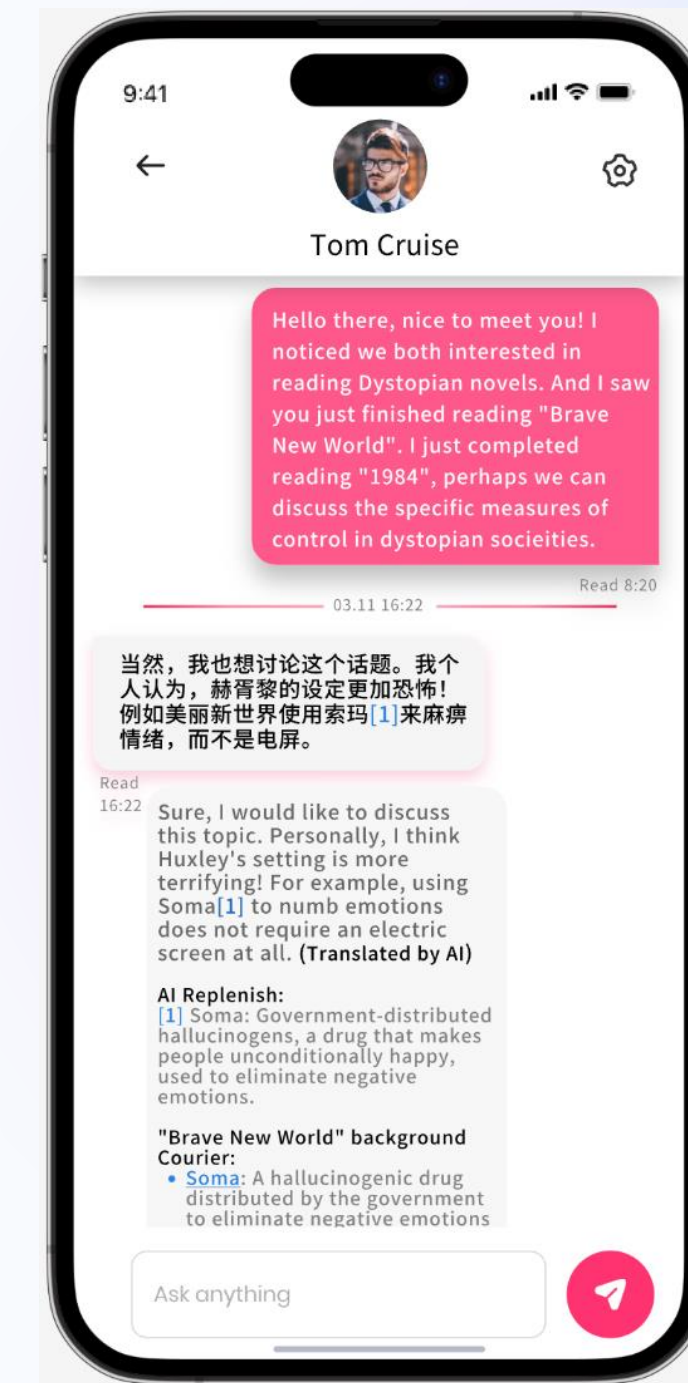
65% of users felt that dialogue reports and derived questions helped to continue the discussion and add depth to the exchange.

Reflection:

- ❑ GAI-powered conversations could lead to over-reliance on AI assistance.

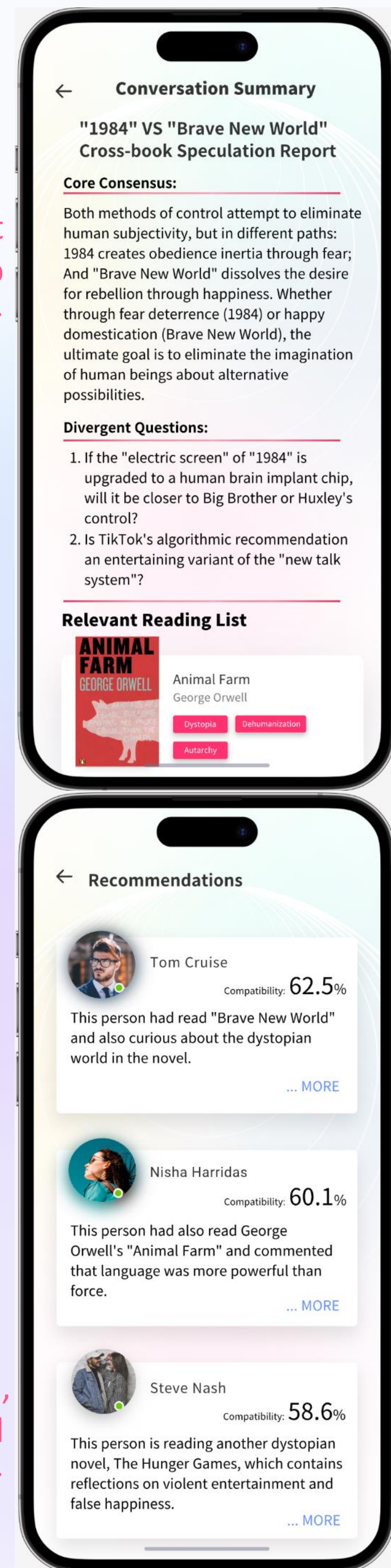
Prototype

Dialogue report
and follow-up
questions→



↑AI generate context-aware translations

Partner selection page,
showing view and
compatibility→



Evaluation Method

To evaluate Readar’s usability and user experience, we used three complementary methods:

- System Usability Scale (SUS)
- Heuristic Evaluation
- A/B Testing

System Usability Scale (SUS=89.1)

What it measures: Ease of learning, efficiency of operation, and overall interface.

Procedure: Each participant (N=21) completed the standard ten-item SUS questionnaire after using Readar for 15 minutes. Using the Likert 5-point scale.

Question	Average Score
Ease of Use	4.67
Intuitive Function Design	4.57
Clear Interface and Natural Operation Logic	4.48
Few Errors and Easy Problem-solving	4.52
Infrequent Discomfort from Prolonged Use	1.38
Coherent Function Design and Natural Module Integration	4.48
Quick Recall of Operations After Long Break	4.62
Minimal Learning Time Required	1.48
Ease of Learning for Most Users	4.52
Overall Satisfaction and Willingness to Recommend	4.62
Total Average	3.93 (std=1.3)

Heuristic Evaluation

What it measures:

- **Consistency:** Overall consistent, but some menu behaviors lack uniformity.
- **Feedback:** Clear in most cases, though some error messages need clearer guidance.
- **Efficiency:** AI features enhance engagement, but recommendation accuracy needs improvement.

Procedure: Three UX experts independently reviewed key workflows and then convened to consolidate findings.

A/B Testing

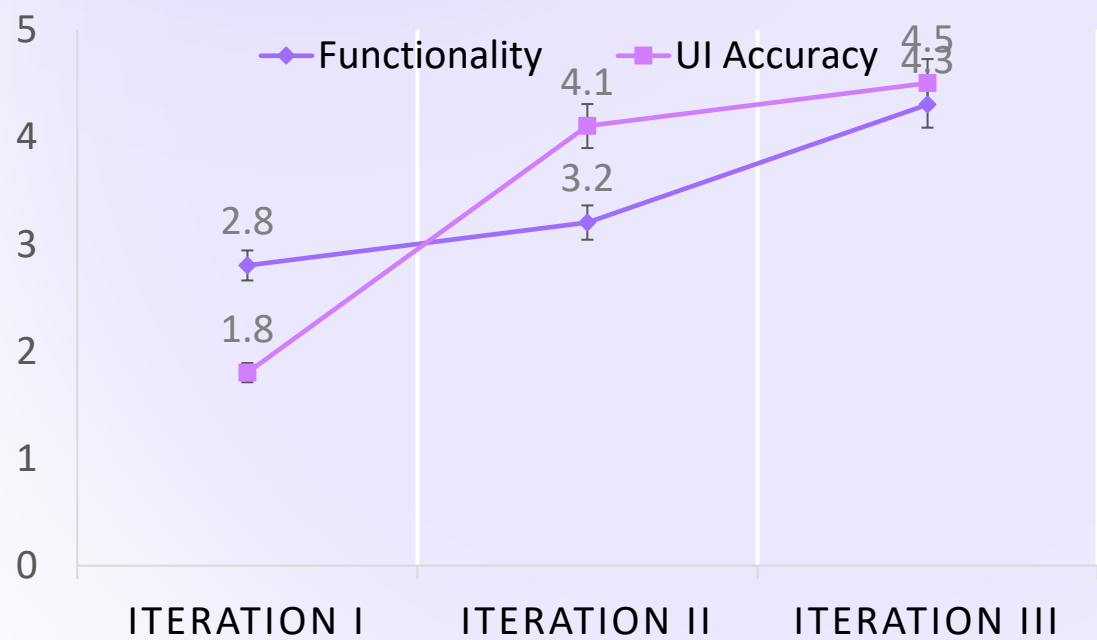
What it measures: Comparative performance of two interface variants on task completion and perceived intuitiveness.

Procedure: Participants were randomly assigned to the control (original navigation) or variant (updated navigation and recommendation layout) and asked to complete a set of reading-discussion tasks.

Impact

Readar significantly improves accessibility and user experience by reducing environmental and technical barriers. Users can generally master the core functions of Readar in a relatively short period of time. Users of different age groups have a high acceptance of Readar. Its intuitive interface and AI assistance effectively relieve pressure, promote deeper thinking, and facilitate inclusive knowledge exchange. The A/B testing also led to improved navigation and a new recommendation layout that better meets user needs.

A/B TESTING



Conclusion

This project demonstrates that AI-assisted, interest-based one-on-one reading discussions can markedly improve depth, engagement, and cross-cultural exchange. While we achieved high usability and match rates, challenges remain around cold-start coverage, subtle cultural translation, and balancing AI support with user flow.

Alignment with Initial Goals

- ✓ Enable depth and structure in post-reading discussions.
- ✓ Facilitate honest and diverse opinions.
- ✓ Match readers by true intellectual affinity rather than mere shared titles.

Future Work

- Further optimize the notification board prompting strategy to reduce disruption to the user's pace of discussion.
- GAI-powered conversations could lead to over-reliance on AI assistance.
- Incorporate voice annotations and gesture recognition to enrich the reading/ discussion experience.

Acknowledgement

I would like to express my thanks to **Prof. Li** and **Prof. Ma** for their insightful teaching and guidance throughout this course. Their support has been invaluable to the development of this project.

I was primarily responsible for the **project initiation, background, servy design, and each iteration process**, as well as the **final evaluation**. This experience has been both challenging and rewarding, and I deeply appreciate the opportunity to explore and contribute to human-centered design.

References

- [1] R. Wu, 'Findings and Analyses from the Readar Questionnaire - CPT208G42', Xi'an Jiaotong-Liverpool University, Suzhou, China, 2025. [Online]. Available: <https://a5eanegz59.feishu.cn/docx/KT0XdadeOoQn99xI35dc4uM2nSf>
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- [3] Cheers Publishing, '《2024职场人阅读报告》 [in Chinese]'. Accessed: Apr. 06, 2025. [Online]. Available: <https://mp.weixin.qq.com/s/UfMUC0-HnQmGLLYGre2raA>

